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DSC680 – Applied Data Science
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# Project 1

## Milestone 1

For my inaugural project, I have selected to employ binary classification to predict fake news. There are numerous fake news datasets available, but I will utilize the dataset from Kaggle: https://www.kaggle.com/datasets/algord/fake-news. This dataset comprises multiple article titles, tweet IDs, and article URLs. Notably, this dataset has been downloaded the most on Kaggle. Given its high number of downloads, we can confidently infer that there has already been community criticism and utilization of this dataset, unlike some datasets that have not undergone proper scrutiny. Consequently, we can conclude that the data is of high quality.

The primary objective of the project is to ascertain the accuracy of predicting fake news with a 75% success rate. Achieving a 75% accuracy rate surpasses the 50/50 guessing threshold, indicating the presence of a significant finding. While there is no assurance that fake news can be definitively distinguished from real news, a contingency plan involves conducting exploratory data analysis (EDA) and conducting storytelling to identify the key features that predominantly contribute to predicting fake news.

Several ethical considerations arise when addressing the issue of fake news. The concept of “freedom of speech” frequently becomes a contentious topic, as its successful implementation could potentially lead to censorship of fake news. However, it is crucial to recognize that fake news is often characterized by spam articles. Consequently, censoring or removing these articles would not constitute an infringement on individual rights, but

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rather a targeted action against the media outlets responsible for producing such content. Furthermore, there exists a moral argument in favor of silencing or eliminating fake news, as it fosters distrust, conspiracies, hatred, confusion, paranoia, and other societal ills. It is highly probable that fake news will have a significant impact on elections or incite unrest.

Initially, we will embark on an exploratory data analysis (EDA) to gain a comprehensive understanding of the data. EDA will serve as a guiding tool in our search while simultaneously elucidating the inherent structure and narrative of the data. Subsequently, we will engage in feature engineering, which involves deriving new features from existing ones. Following this, we will partition the data into training and test sets, and subsequently, we will evaluate the classification metrics to ascertain whether our objectives were met. Subsequently, hyperparameter tuning and fine-tuning will be conducted to enhance the precision of our model. Finally, irrespective of the outcomes, we will present a narrative conclusion that elucidates our proposed solutions and elucidates any insights gained.